

Control of a quadruple tank system 01

Consider the quadruple tank system described in the attached paper. Overall, the linearized system dynamics (around a given equilibrium point) can be described by the following model:

$$\dot{x} = Ax + B_1u_1 + B_2u_2$$

$$y_i = C_i x, i = 1, 2$$

where the system matrices are specified in the file.

Problem:

1. Generate the system matrices (both continuous-time and discrete-time, the latter with a sampling time of choice, selected according to the system dynamics). Perform the following analysis:
 - a. Compute the eigenvalues and the spectral abscissa of the (continuous-time) system. Is it open-loop asymptotically stable?
 - b. Compute the eigenvalues and the spectral radius of the (discrete-time) system. Is it open-loop asymptotically stable?
2. For different control structures (i.e., centralized, decentralized, and different distributed schemes) perform the following actions
 - a. Compute the continuous-time fixed modes
 - b. Compute the discrete-time fixed modes
 - c. Compute, if possible, the CONTINUOUS-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
 - d. Compute, if possible, the DISCRETE-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
 - e. Analyze the properties of the so-obtained closed-loop systems (e.g., stability, eigenvalues) and compute the closed-loop system trajectories (generated both in continuous-time and in discrete-time) of the system variables, starting from a common random initial condition.